Course Description and Outcome Form
Department of Electrical and Electronic Engineering
School of Engineering
Brac University

EEE 400/ECE 402 Final Year Design Project

A. Course General Information:

Course Code and Title and Credit Hours	Contact Hours/Week	Broader Goals	Semester
EEE 400P Final Year Design Project (2 credit hours) / ECE 402P Final Year Design Project (1.5 credit hours)	3	Problem Identification and Project Proposal	Semester 1 of final year design project
EEE 400D Final Year Design Project (2 credit hours) / ECE 402D Final Year Design Project (1.5 credit hours)	3	Design and Development	Semester 2 of final year design project
EEE 400C Final Year Design Project (2 credit hours) / ECE 402C Final Year Design Project (1.5 credit hours)	3	Validation and Project Completion	Semester 3 of final year design project

Category:	Program Core
Type:	Required, Engineering
Prerequisites (EEE400):	MGT 211 Principles of Management ACT 201 Financial Accounting EEE 223 Energy Conversion II EEE 224 Energy Conversion Laboratory EEE 305 Control Systems EEE 306 Control Systems Laboratory EEE 341 Introduction to Communication Engineering EEE 342 Introduction to Communication Engineering Laboratory EEE 343 Digital Signal Processing EEE 344 Digital Signal Processing Laboratory EEE 365 Microprocessor and Interface EEE 366 Microprocessor and Interface Laboratory
Prerequisites (ECE402):	MGT 211 Principles of Management ACT 201 Financial Accounting ECE 343 Digital Signal Processing ECE 344 Digital Signal Processing Laboratory EEE 365 Microprocessor and Interface EEE 366 Microprocessor and Interface Laboratory ECE 411 VLSI Design ECE 412 VLSI Design Laboratory (ECE 441 Wireless and Mobile Communications / ECE 445 Digital Communication) (ECE 442 Wireless and Mobile communication Laboratory / ECE 446 Digital Communication Laboratory)
Co-requisites:	None
Credit requirements:	At least 100 credit hours completed

B. Course Catalog Description (Content):

The Final Year Design Project (FYDP) is the first step towards transferring students' experience from the academic environment to the industry. The course provides a culminating assessment of the students by applying and integrating their previously acquired knowledge to the solution of complex electrical and electronic engineering problem. The primary focus of the Final Year Design Project is to improve the students' technical skills, communication skills and teamwork opportunities through an electrical and electronic engineering project development work. It also focuses on variety of non-technical issues such as professional and ethical responsibilities and practices, safety, reliability, legal cultural, social and environmental impacts as well as sustainability of engineering solution

The Final Year Design Project course consists of two parts: Instructional Part and Technical Demonstration Part. The major topics covered in the Instructional Part include:

- Overview of the Final Year Design Project course, student learning outcomes, expectation, assessment, checklist etc.
- Introduction to engineering design process including formulation of problem, analysis of objectives, specifications and requirements, consideration of realistic constraints, engineering standards and impact of engineering solutions, design of solution, implementation, evaluation and validation of the solution
- Review of project proposal preparation, estimating, project management and scheduling etc.
- Review of engineering ethics and professional practices
- Safety in engineering design.
- Contemporary issues and life-long learning
- Report writing and presentation techniques
- Teamwork building

The Technical Demonstration part primarily includes various activities including (but not limited to):

- Literature review and research
- Identification and formulation of project problem
- Analysis of objectives, specifications and requirements,
- Project plan, proposal and management
- Implementation of design process
- Design reviews, simulation and finalization
- Development of solution, testing and validation
- Documentation, drawings, written reports, oral presentation etc.

C. Course Objectives:

The objectives of the Final Year Design Project are to:

a. Provide students opportunity to apply and integrate their previously acquired engineering knowledge to the solution of engineering problem
b. Enhance student's creativity in analyzing and solving complex and possibly real-world engineering problems.
c. Train students with skills on systematic design and development process and documentation to the solution of engineering project
d. Prepare student to develop and enhance self-learning ability.
e. Prepare students experience of engineering project development that will be useful in their industrial careers.
f. Aware students regarding professional practices, norms and ethical responsibilities in regards to designing engineering solution
g. Prepare student to understand and evaluate the impact of engineering solutions to the society, health, safety, reliability, legal, cultural social
h. Prepare students to understand and evaluate the sustainability and impact of engineering solution towards environment
i. Create an environment to promote team approach in engineering problem solving
j. Develop communication skill among students through complex activities, technical report writing, oral presentations etc.

D. General Guideline:

- **a.** The EEE 400/ ECE 402 Final Year Design Project course will be offered in 3 consecutive semesters of student's final year of the program.
- **b.** Final Year Design Project (FYDP) Coordinator Committee – will manage and coordinate overall activities, instructional part, team formation, assigning Academic Technical Committee to each team, arranging progress presentation sessions in regular basis, assessments, Project Showcase event etc.
- **c.** Academic Technical Committee - The Academic Technical Committee (ATC) will consist of 1 faculty advisor and minimum 2 members from faculty and/or industry.
- **d.** The faculty advisor will be the chair of the ATC.
- e. ATC will focus more on project methodology and guide the students accordingly, assess and evaluate students' performance and learning outcomes, prepare CO assessment reports, submit grades.
- f. Students will work in team of no more than 4 members to design and develop the solution of a complex engineering problem.
- g. Each team will be assigned to one Academic Technical Committee (ATC).
- h. Team will meet with the ATC in regular basis to discuss and update project progress and issues.
- i. Team will do most of the work at their own
- j. Team members will share any cost related to project development.
- k. Team will submit deliverables, presentations at various stages of the project

E. Semester-wise Distribution of Tasks and Responsibilities:

The various task and responsibilities of the year-long EEE 400 Final Year Design Project course will be distributed in 3 consecutive semesters as following manner:

EEE 400P/ECE 402P:

Semester 1 (this is the first of three consecutive semesters of FYDP course)	
Course Code and Title Credit Hours:	EEE 400P Final Year Design Project (2 credit hours) / ECE 402P Final Year Design Project (1.5 credit hours)
Contact Hours	3 hours/week
Purpose	Problem Identification, Project Planning and Proposal
Tasks and Responsibilities:	*Responsibility of students* - Students participate in Instruction part in assigned time slot and venue (attendance will be taken) - Team meets assigned ATC in regular basis to discuss about project technical issues and progress - Team will do most of the work at their own times - Team presents project progress in assigned time slot and venue (attendance will be taken) - Team submits Project Concept Note to ATC by the middle of the semester - Team submits Project Proposal Report to ATC at the end of the semester - Individual members submit Peer Evaluation Form at the end of the semester

EEE 400D/ECE 402D:

Semester 2 (this is the 2nd of three consecutive semesters of FYDP course)	
Course Code and Title Credit Hours:	EEE 400D Final Year Design Project (2 credit hours) / ECE 402D Final Year Design Project (1.5 credit hours)
Contact Hours:	3 hours/week
Purpose :	Design and Develop of the Solutions
Tasks and Responsibilities:	*Responsibility of students* - Students participate in instruction part, if any, in assigned time slot and venue (attendance will be taken) - Team meets respective ATC in regular basis to discuss about project issues and progress - Team submits progress report to the ATC in the middle of the semester - Team presents project progress on working prototype in assigned time slot and venue scheduled by Coordination Committee - Team submits Design Report to ATC at the end of the semester - Individual members submit Peer Evaluation Form at the end of the semester

EEE 400C/ECE 402C:

Semester 3 (this is the last of three consecutive semesters of FYDP course)	
Course Code and Title Credit Hours:	EEE 400C Final Year Design Project (2 credit hours) / ECE 402C Final Year Design Project (1.5 credit hours)
Contact Hours	3 hours/week
Major Purpose:	Project Completion
Tasks and Responsibilities:	*Responsibility of students* - Team meets respective ATC in regular basis to discuss about project issues and progress - Team presents project progress in assigned time slot and venue - Team submit Progress Report on working prototype development to ATC - Team Demonstrate working prototype to ATC in the middle of the semester - Team submit Project Final Report to ATC by the end of the semester - Team deliver Final Presentation as scheduled by Coordination Committee - Team participate in Final Year Design Project Showcase to demonstrate the final prototype/solution organized by Coordination Committee - Individual members submit Peer Evaluation Form

F. CO Assessment Plan, Activities, Deliverables and Timeline

EEE 400P/ECE 402P:

Semester 1 of FYDP (10th Semester of the Program)			
Sl.	**CO Description**	**Activities**	**Assessment Tools**
CO1	**Identify** a solvable complex engineering problem preferably relevant to the current and future industry through appropriate research.	Review of literature, background study to identify a problem to solve Prepare problem statement with scope and justification	• Submission of Project Concept Note (7th Week)
CO2	**Identify** the objectives, specifications, functional and non-functional requirements, and constraints as well as applicable compliance, standards and codes of practice to the solution of the engineering problem.	Feedback from stakeholder to find out objectives, specifications and functional requirements Review of applicable standards, professional codes, practice and regulatory requirements	
CO3	**Assess** the impact of the solution of the engineering project in terms of societal, health, safety, legal and cultural context.	Study of project impact on society, health and safety, legal and culture Write a report on the impact	• Submission of Project Proposal Report (12th Week)
CO4	**Evaluate** the sustainability and impact of solution of the proposed project in terms of environmental consideration.	Study of project sustainability and impact on environment Write a report on the sustainability and impact	
CO11	**Demonstrate** project management skill in various stages of developing the solution of engineering design project.	Obtain the final specifications and requirements Determine appropriate analysis and design methodology Identify and assess required and applicable resources Identify project risk issues and how to address that. Prepare a complete project management plan including schedule, budget, resource and risk management	• Submission of Project Proposal Report (12th Week) • Progress Presentations (4th, 8th , 12th Week)
CO13	**Apply** ethical considerations and professional responsibilities in designing the solution and throughout the project development phases.	Identify ethical issues and professional responsibility related to design solution	• Submission of Peer-evaluation form (12th Week) • Instructor's Assessment
CO14	**Perform** effectively as an individual and as a team member for successfully completion of the project.	Active participation in team activities Complete peer-evaluation form	

CO15	**Communicate** effectively through writings, journals, technical reports, deliverables, presentations and verbal communication as appropriate at various stages of project development.	Verbal and written communication within the team and appropriate stakeholders Write notes, Journals Prepare problem concept notes, project proposal reports Prepare presentations Deliver Oral presentation	● Submission of Project Concept Note (7th Week) ● Submission of Project Proposal Report (12th Week) ● Progress Presentations (4th, 8th , 12th Week)

EEE 400D/ECE 402D:

Semester 2 (11th Semester of the Program)			
Sl.	**CO Description**	**Activities**	**Assessment Tools**
CO5	**Design** multiple engineering solutions of the problem to meet the desired objectives, need and requirements within the given constraints.	Preparing design process as per objectives, specifications, requirements and constraints Preliminary design of multiple alternative solutions Perform simulation of the alternative design solutions for functional verification	● Submission of Design report (12th Week)
CO6	**Analyze** alternative design solutions of engineering problems in order to find the most appropriate one considering cost, efficiency, usability, manufacturability, impact, sustainability, maintainability etc.	Perform analysis of alternative design solutions to obtained the best suitable solution Perform optimization of the selected design solution	
CO9	**Use** modern engineering and IT tools to design, develop and validate the solution.	Selection of appropriate tools to design and analyze the solutions Usage of tools as necessary	● Submission of Design report (12th Week)
CO10	**Conduct** independent research, literature survey and learning of new technologies and concepts as appropriate to design, develop and validate the solution.	Find out appropriate knowledge information, skill, materials, and resources as needed to design and analyze the solutions Use the learned knowledge and skills as needed	
CO11	**Demonstrate** project management skill in various stages of developing the solution of engineering design project.	Evaluate project progress against the project plan Prepare revised project plan and reschedule as required to complete the project within the stipulated time	● Submission of Design report (12th Week) ● Project Progress Presentations (4th, 8th , 12th Week)
CO13	**Apply** ethical considerations and professional responsibilities in designing the solution and throughout the project development phases.	Identify ethical issues and professional responsibility related to design solution	● Submission of Peer-evaluation form (12th Week) ● Instructor's Assessment

CO14	**Perform** effectively as an individual and as a team member for successfully completion of the project.	Active participation in team activities Complete peer-evaluation form	
CO15	**Communicate** effectively through writings, journals, technical reports, deliverables, presentations and verbal communication as appropriate at various stages of project development.	Verbal and written communication within the team and appropriate stakeholders Write notes, Journals Prepare design reports Prepare presentations Deliver Oral Presentation	• Submission of Design report (12th Week) • Project Progress Presentations (4th, 8th , 12th Week)

EEE 400C/ECE 402C:

Semester 3 (12th Semester of the Program) **EEE 400C Final Year Design Project**			
Sl.	**CO Description**	**Activities**	**Assessment Tools**
CO7	**Evaluate** the performance of the developed solution with respect to the given specifications, requirements and standards.	Implement the selected design solution Perform testing of the Implemented solution Evaluate the performance of the implemented solution	• Submission of Progress report on working prototype development (7th Week) • Demonstration of working prototype (7th Week)
CO8	**Complete** the final design and development of the solution with necessary adjustment based on performance evaluation.	Obtain the final design by make necessary adjustment to the selected design as per performance evaluation result Develop the solution as per final design	• Submission of Project Final Report (12th Week) • Final Presentation (12th Week or later as scheduled by Coordination Committee) • Demonstration of completed prototype in Final Year Design Project Showcase event organized by department
CO9	**Use** modern engineering and IT tools to design , develop and validate the solution.	Selection of appropriate tools to develop, and validate the selected solution and final prototype Usage of tools as necessary	• Submission of Project Final Report (12th Week)
CO10	**Conduct** independent research, literature survey and learning of new technologies and concepts as appropriate to design, develop and validate the solution.	Find out appropriate knowledge information, skill, materials, and resources as needed to develop and validate the final prototype Use the learned knowledge and skills as needed	
CO11	**Demonstrate** project management skill in various stages of developing the solution of engineering design project.	Evaluate project progress as per schedule Prepare revised project plan and reschedule as required to	

		complete the project within the stipulated time	
CO12	**Perform** cost-benefit and economic analysis of the solution.	Conduct economic analysis Conduct cost-benefit estimation	
CO13	**Apply** ethical considerations and professional responsibilities in designing the solution and throughout the project development phases.	Identify ethical issues and professional responsibility related to design solution	• Submission of Peer-evaluation form (12th Week) • Instructor's Assessment • Submission of Project Final Report (12th Week)
CO14	**Perform** effectively as an individual and as a team member for successfully completion of the project.	Active participation in team activities Complete peer-evaluation form	• Submission of Peer-evaluation form (12th Week) • Instructor's Assessment
CO15	**Communicate** effectively through writings, journals, technical reports, deliverables, presentations and verbal communication as appropriate at various stages of project development.	Verbal and written communication within the team and appropriate stakeholders Write notes, Journals Prepare project final report Prepare presentations Deliver Oral presentations Deliver final demonstration of the completed prototype	• Submission of Project Final Report (12th Week) • Final Presentation (12th Week or later as scheduled by Coordination Committee) • Demonstration of completed prototype in Final Year Design Project Showcase event organized by department

G. Distribution of Assessment Points

	Distribution of assessment points among various COs assessed in different semesters														
PO	**l**	**c**	**f**	**g**	**c**	**b**	**d**	**c**	**e**	**l**	**k**	**k**	**h**	**i**	**j**
CO	**CO 1**	**CO 2**	**CO 3**	**CO 4**	**CO 5**	**CO 6**	**CO 7**	**CO 8**	**CO 9**	**CO 10**	**CO 11**	**CO 12**	**CO 13**	**CO 14**	**CO 15**
EEE 400P/ ECE 402P (out of 100)	15	21	15	15							15		6	6	7
EEE 400D/ ECE 402D (out of 100)					30	30			15	5	5		5	5	5
EEE 400C/ ECE 402C (out of 100)							30	24	6	4	4	6	7	7	12
Total Points (300)	15	21	15	15	30	30	30	24	21	9	24	6	18	18	24
Weightage (% of Total Points)	5%	7%	5%	5%	10 %	10 %	10 %	8%	7%	3%	8%	2%	6%	6%	8%

Weightage among various POs assessed in FYDP																
PO	**b**	**c**			**d**	**e**	**f**	**g**	**h**	**i**	**j**	**k**		**l**		
CO	CO 6	CO 2	CO 5	CO 8	CO 7	CO 9	CO 3	CO 4	CO 13	CO 14	CO 15	CO 11	CO 12	CO 1	CO 10	
Weightage	10%	25%			10%	7%	5%	5%	6%	6%	8%	10%		8%		

H. Tentative Class time for Instructional part and Progress Presentation:

Will be provided as attachment

I. Grading Policy

a. Brac University grading policy will be applied to determine the letter grade of the individual student

J. CO Attainment Policy:

a. Points earned in all three semesters (400P/402P, 400D/402D and 400C/402C) will be used to determine the student's overall performance in CO assessment as per the CO assessment plan
b. The CO attainment policy will be as per the course outcome attainment policy of the Department of Electrical and Electronic Engineering

K. Guideline on Report Format and Presentation

a. Supplementary materials will be provided.